Analysis and Optimization of Pseudo-Boolean Functions using the Fast Walsh-Hadamard Transform

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1 Introduction and Scope

Context and Motivation: [1] In numerous combinatorial optimization and computational biology domains, complex problems are mathematically modeled as pseudo-boolean functions, defined as mappings from an n-dimensional binary vector space to the set of real numbers:

$$f : \{0,1\}^n \to \mathbb{R} \tag{1}$$

In real-world applications, these functions often behave as strict "black-boxes," meaning their internal algebraic structures, dependencies, and landscapes are entirely unknown prior to evaluation.

- **The Combinatorial Bottleneck:** The primary challenge in black-box optimization is the combinatorial explosion of the search space. For n binary variables, there are 2^n unique combinations. Evaluating every point via naive search becomes computationally intractable even for moderate values of n.
- Greedy algorithms frequently fail in these spaces because variables are often interconnected; optimizing one variable locally may degrade the overall global fitness.

Scope of the Report: To overcome these limitations, this report introduces a framework based on Walsh decomposition to mathematically "look inside" the black box. The scope covers the definition of the Walsh transform, the algorithmic acceleration achieved via the Fast Walsh-Hadamard Transform (FWHT), and the evaluation of this framework using NK-Landscapes. Finally, we provide an empirical analysis of Walsh coefficient distributions, demonstrating how they uncover hidden linkages and enable high-speed surrogate modeling.

2 Definition of the Walsh Transform and Variants

2.1 The Walsh Polynomial

[2] The Walsh transform acts as a discrete counterpart to the Fourier transform for binary spaces. It projects any pseudo-boolean function $f(x)$ onto a complete, orthogonal basis of Walsh functions. The algebraic spectral form is uniquely represented by the Walsh Polynomial:

$$f(x) = \sum_{k=0}^{2^n-1} w_k \cdot \varphi_k(x) \tag{2}$$

- n: The number of binary variables.
- w_k: The Walsh coefficient corresponding to the spectral index k.
- $\varphi_k(x)$: The k-th Walsh function, which outputs $+1$ or -1 based on the parity of the active bits defined by the mask k.

2.2 Physical Meaning of the Coefficients

[3] The coefficients (w_k) quantify the hidden structure of the black box.

- w_0 represents the global average of the entire search space.
- **First-order coefficients** (where k has a Hamming weight of 1) measure the independent linear contribution of a single variable.
- **High-order coefficients** (Hamming weight ≥ 2) explicitly quantify the non-linear interaction (epistasis) between groups of variables.

2.3 Standard Matrix Derivation vs. Fast Walsh-Hadamard Transform (FWHT)

Traditionally, the full vector of coefficients $\underline{w}$ is derived using the symmetric Hadamard Matrix (H_n), constructed recursively via Sylvester's method:

$$\underline{w} = \frac{1}{2^n} \cdot H_n \cdot f \tag{3}$$

The matrix H_n of dimension $2^n \times 2^n$ is generated recursively using Sylvester's construction method:

$$H_0 = [1] \tag{4}$$

$$H_n = \begin{bmatrix} H_{n-1} & H_{n-1} \\ H_{n-1} & -H_{n-1} \end{bmatrix} \tag{5}$$

However, multiplying a $2^n \times 2^n$ matrix requires $O(2^{2n})$ time complexity, which scales exponentially squared. To solve this, the Fast Walsh-Hadamard Transform (FWHT) applies a divide-and-conquer strategy. [4] By exploiting the recursive Kronecker structure of the Hadamard matrix, the FWHT processes the array in n stages using pairs of additions and subtractions known as the "Butterfly Operation". This reduces the time complexity drastically to $O(n2^n)$ and allows for in-place memory calculation.

2.4 Variant: Localized Micro-FWHT for Massive Scale (Big N)

When N scales to massive dimensions (e.g., $N = 1000$), even the accelerated $O(n2^n)$ complexity of the standard FWHT completely fails due to the 2^n combinatorial explosion. However, for a specific class of problems, this memory and processing bottleneck can be elegantly bypassed by exploiting localized mathematical architectures.

It is crucial to emphasize that **the Localized Micro-FWHT is not a general-purpose method applicable to all pseudo-boolean black-box functions**. In a generic black-box scenario, one only has oracle access to the global function value $f(x)$ and cannot decompose the system into independent components. The Localized Micro-FWHT instead requires a "grey-box" access pattern, which is perfectly satisfied by the **NK-Landscape** model.

The NK-Landscape is defined as the average of N localized, explicitly visible sub-functions, where each sub-function relies strictly on its core variable and K epistatic neighbors, totaling exactly $K + 1$ variables:

$$f(x) = \frac{1}{N} \sum_{i=1}^{N} f_i(x_i, z_{i,1}, z_{i,2}, \ldots, z_{i,K}) \tag{6}$$

Because of this explicit, decomposable structural constraint, we can apply the Localized Micro-FWHT. Instead of attempting to transform the intractable global 2^n space, the algorithm directly accesses the internal representation of the NK-Landscape to extract the localized lookup tables (or truth table vectors) of size 2^{K+1} for each sub-function f_i. It then executes a standard, localized micro-FWHT on each of these N sub-spaces individually, scaling within a minimal $O((K+1)\cdot 2^{K+1})$ complexity per sub-function.

$$w_k = \frac{1}{N}\sum_{i=1}^{N} w_k^{(i)} \tag{7}$$

The resulting local coefficients $w_k^{(i)}$ are then shifted to align with their global variable indices, multiplied by the normalization factor of $\frac{1}{N}$, and accumulated into the global spectrum. To prevent out-of-memory errors, this projection heavily utilizes Sparse Data Structures (such as HashMaps). Because the maximum number of non-zero Walsh coefficients in an NK-Landscape is mathematically bounded by $N\cdot 2^{K+1}$, this sparse mapping drops the total algorithmic complexity to a highly tractable $O(N\cdot K\cdot 2^K)$, rendering massive-scale analysis instantaneous.

3 Definition of the Benchmark Functions: The NK-Landscapes

3.1 Internal Mechanics and Lookup Tables

As introduced in the Micro-FWHT variant, the NK-Landscape is defined by N variables and K interacting neighbors. From an algorithmic perspective, the landscape is constructed using two core structures: the interaction map and the local fitness lookup tables.

For every variable x_i, an interaction array randomly assigns K other variables that influence its state. [5] Consequently, a lookup table of size 2^{K+1} is generated for each variable, populated with random real numbers drawn from a uniform distribution $U(0,1)$. When the function is evaluated, the binary states of x_i and its K neighbors act as an index to retrieve a specific fitness value from this table.

3.2 Epistasis, Frustration, and Ruggedness

The NK-Landscape serves as an exceptional benchmark because it allows researchers to explicitly control the degree of "epistasis" [5] (non-linear variable linkage). When $K=0$, the variables are completely independent, and the landscape is smooth and linear.

However, as K increases, a single variable participates in multiple sub-functions simultaneously. Changing the state of one bit to optimize a local sub-function inevitably alters the lookup table indices of interacting sub-functions, potentially degrading their fitness values. This phenomenon, known as "frustration," transforms the search space into a highly rugged topography filled with deceptive local optima, defeating standard greedy algorithms.

3.3 Theoretical Sparsity

Despite this ruggedness, the NK-Landscape possesses a profound mathematical property that makes it ideal for Walsh analysis: extreme sparsity. [6] Because the epistatic interactions are strictly localized to K neighbors, the maximum Walsh interaction order is rigorously bounded by $K+1$. Furthermore, the total number of non-zero Walsh coefficients is mathematically capped at $N\cdot 2^{K+1}$. This guarantees that even in an astronomically large 2^N search space, the underlying algebraic structure remains highly compressed and can be fully decoded.

4 Experimental Results: Coefficient Distribution and Importance

4.1 Coefficient Distribution Analysis ($n = 1000, k = 2$)

The Walsh coefficient distributions across orders 1–3 exhibit exponential decay, with coefficients heavily concentrated at lower magnitudes. Order 1 dominates with the highest frequency of significant terms, while higher orders contribute progressively less. This pattern indicates that the NK landscape structure is primarily determined by lower-order interactions, suggesting that first-order terms capture the essential landscape complexity and enabling reasonable approximation of the fitness surface using fewer Walsh basis functions.

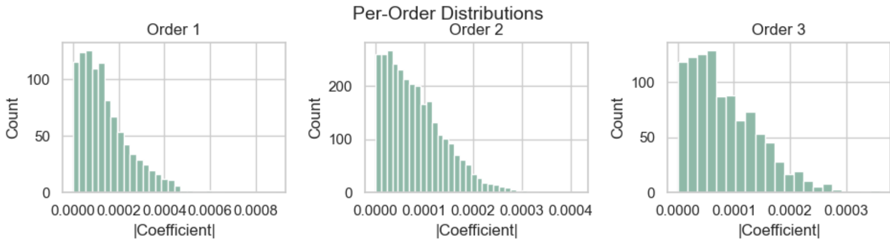

Figure 1: Walsh coefficient magnitude distribution across orders 1–3 for $n = 1000, k = 2$, demonstrating exponential decay.

4.2 Parameter Scaling Dynamics: Problem Size (n) vs. Epistasis Strength (k)

We analyze how problem size (n) and epistasis strength (k) scale landscape complexity, revealing a stark mathematical asymmetry:

- **Epistasis Scaling (k):** Increasing k from 3 to 4 (with $n = 1000$) triggers an exponential explosion in non-zero coefficients (+125%, $11,967 \rightarrow 26,929$), while their mean magnitude decays by 41% ($1.04 \times 10^{-4} \rightarrow 6.12 \times 10^{-5}$). Higher epistasis generates numerous weak, highly-distributed interactions, increasing topological ruggedness.
- **Problem Size Scaling (n):** Doubling n from 500 to 1000 (with $k = 3$) causes a strictly linear doubling of coefficients ($5,979 \rightarrow 11,967$) and a proportional 50% decrease in mean magnitude ($2.10 \times 10^{-4} \rightarrow 1.04 \times 10^{-4}$), validating the theoretical $N \cdot 2^{K+1}$ sparsity bound without increasing topological ruggedness.

This linear-vs-exponential scaling confirms that optimization algorithms must prioritize discovering linkage groups (handling k) over merely managing structural scale (handling n).

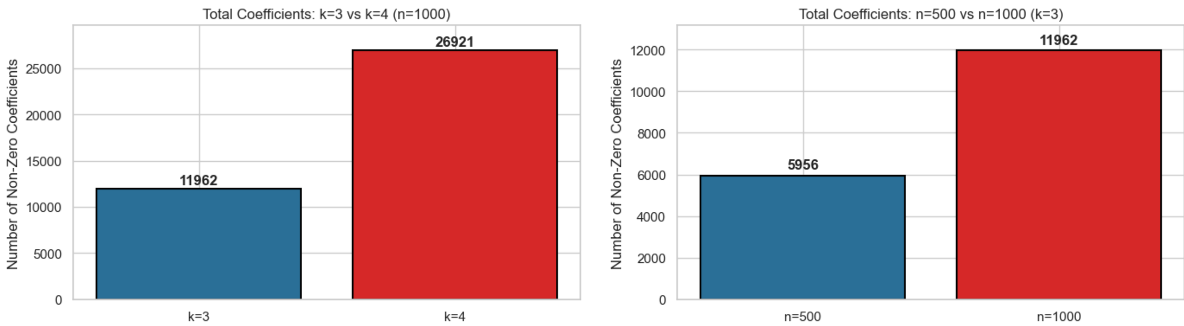

Figure 2: Walsh coefficient distributions across interaction orders, representing the same scaling analysis plotted under different parameter configurations (n, k) .

4.3 Mean Density Analysis and Bound Tightness

To evaluate the tightness of the theoretical $N \cdot 2^{K+1}$ sparsity bound, we analyze the statistical density of Walsh coefficients across 20 random NK-landscape instances for both $k = 3$ and $k = 4$ configurations (with $n = 1000$ fixed). Specifically, the maximum Walsh coefficient limit for any degree $d > 1$ is given by the formula:

$$\text{Limit}_{d>1} = n \times \binom{k+1}{d} \tag{8}$$

where for degree $d = 1$, the limit is strictly n. The empirical coefficient densities, mean counts, and standard deviations for each interaction order are summarized side-by-side in Table 1 for a concise comparison.

Table 1: Walsh Coefficient Density Statistics Across 20 Instances ($n = 1000$) for $k = 3$ and $k = 4$.

	Epistasis Strength $k = 3$				**Epistasis Strength $k = 4$**			
Order (d)	**Max**	**Mean**	**S.D.**	**Density**	**Max**	**Mean**	**S.D.**	**Density**
1	1,000	1,000.0	0.00	100.00%	1,000	1,000.0	0.00	100.00%
2	6,000	5,967.7	6.56	99.46%	10,000	9,909.0	9.72	99.09%
3	4,000	3,999.9	0.30	100.00%	10,000	10,000.0	0.22	100.00%
4	1,000	1,000.0	0.00	100.00%	5,000	5,000.0	0.00	100.00%
5		–			1,000	1,000.0	0.00	100.00%

The empirical results reveal that the Walsh spectrum of random NK-landscapes is almost completely dense up to order $K + 1$, with an overall density exceeding 99.9% in both cases. Except for minor stochastic cancellations in second-order interactions (yielding 99.46% density for $k = 3$ and 99.09% density for $k = 4$), all other orders consistently reach exactly 100% capacity with negligible variance. This confirms that the theoretical interaction bound is exceptionally tight, showing that random sub-functions rarely undergo algebraic cancellation in the Walsh basis.

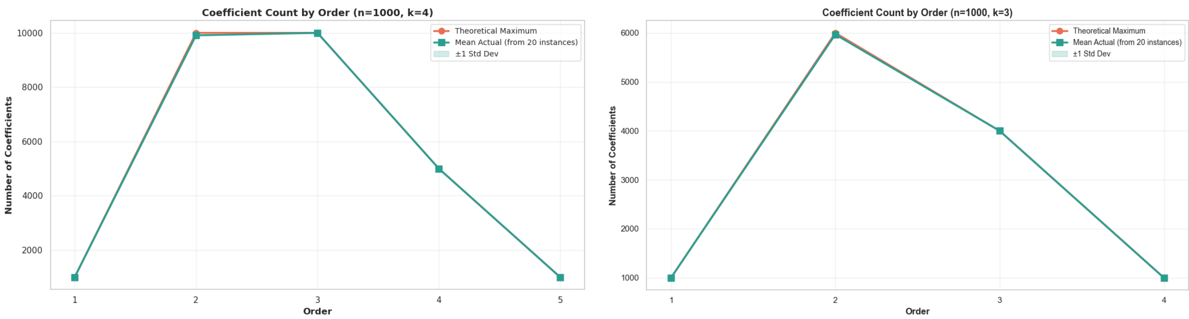

Figure 3: Empirical Walsh coefficient statistics and density analysis across 20 NK-landscape instances, representing the same evaluation of bound tightness plotted under different parameter configurations (n, k) .

4.4 Implications for Black-Box Optimization

The comprehensive empirical and theoretical analysis of Walsh coefficients yields critical insights for black-box optimization on pseudo-Boolean landscapes:

- **Global Sparsity and Bounded Epistasis:** Although a search space of size $n = 1000$ yields an astronomical total of 2^{1000} possible Walsh coefficients, the Walsh spectrum of an NK-landscape is globally extremely sparse. For example, for $k = 4$, only $27,000$ coefficients are non-zero. This mathematical guarantee of bounded epistasis means the entire landscape can be fully characterized by low-order interactions of degree $d \leq k + 1$. Consequently, instead of requiring exponential evaluations, the exact landscape can be reconstructed in polynomial time using sparse recovery techniques or interpolation.

- **High Micro-Level Ruggedness:** The near 100% local density of Walsh coefficients up to order $k + 1$ (with minor stochastic cancellations yielding over 99% density in second-order interactions) implies that almost all possible low-order epistatic interactions are actively engaged. For optimization practitioners, this dense configuration signals high ruggedness at local search scales. Standard local search heuristics (such as hill climbing or simple mutation-based evolutionary algorithms) will inevitably struggle due to an extremely large population of local optima and highly correlated deceptive sub-structures.
- **Leveraging Gray-Box Optimization and Surrogates:** [7] The localized density and global sparsity pattern of Walsh coefficients makes these functions outstanding candidates for gray-box optimization and surrogate-assisted evolutionary algorithms (SAEAs). Instead of treating the objective function as a pure black box, optimization algorithms can exploit the exact Walsh factorization of sub-functions. Techniques like partition crossover or tree-decomposition can analytically identify globally optimal variable assignments directly from the non-zero low-order Walsh coefficients, bypassing the need for exhaustive evaluations.

References

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